Research Brief: Chrome Extension Development

Should you build one? A five-perspective analysis — Practitioner, Skeptic, Economist, Historian, Academic

---

Executive Summary

Building a Chrome extension is cheap and mechanically fast, and it buys you a kind of browser access most software never gets — but that access is borrowed, not owned, and its terms have already been unilaterally rewritten once (Manifest V2→V3) and will likely be rewritten again. The evidence for that historical pattern is solid; the evidence for exactly how severe the next round will be is not. Set against that risk is a plainer, more immediate fact: extensions rarely make money on their own — they're a funnel or an acquisition target — so the real question isn't just "will Google restrict me" but "do I have a monetization path independent of the extension itself." Build one if deep browser integration is genuinely core to your product and you can survive with minimal permissions; don't build one as a distribution shortcut expecting the extension to be the business.

Five Key Findings

#	Finding	Confidence
1	The technical build is low-friction — manifest, service worker, content scripts, zip file. Directly verifiable by anyone who tries it.	9 / 10
2	Chrome has already unilaterally restricted extension capability once, at scale (Manifest V2→V3), following a documented historical pattern also seen in Internet Explorer toolbars and Firefox XUL/XPCOM→WebExtensions. Real precedent, not speculation.	8 / 10
3	Manifest V3 measurably reduced capability for a specific category (ad blockers, privacy tools) via webRequest restrictions. Direction is true and reported; treat exact figures with care unless traced to a primary source.	6 / 10
4	Extensions are rarely a standalone business — more common outcomes are (a) a lead-gen funnel to a paid backend, or (b) an acquisition target whose trusted install base gets repurposed for ads or data collection.	7 / 10
5	Broad permission requests correlate with both attack surface and user-trust erosion over time. Directionally well-supported by published security research (see References).	7 / 10

Weakest claim: *Finding #3, in earlier drafts, leaned on unverified citation texture ("researchers crawled tens of thousands of extensions") without a traceable source. The underlying fact — V3 reduced ad-blocker capability — is confirmed by multiple independent outlets below; the earlier framing simply overstated how precisely that fact had been sourced.*

Scenario-Based Findings: "This Has Happened Before"

Each finding below is paired with a documented historical scenario and the primary or secondary sources it draws on, so the pattern-claims in this brief can be checked rather than taken on faith.

Scenario 1 — Permissive extension model exploited for spyware/adware (Internet Explorer, 1997–2009)

Internet Explorer's Browser Helper Objects (BHOs), introduced in 1997, gave third-party add-ons unrestricted access to the browser's Document Object Model with no review process. This openness was quickly exploited: adware distributors bundled toolbars with ad injection, and BHOs were used to build keyloggers and spyware. Microsoft didn't add a user-facing add-on manager to review/disable BHOs until Internet Explorer 6 Service Pack 2 — nearly a decade after BHOs launched, and only after the abuse pattern was well established.

- Wikipedia, "Browser Helper Object" — en.wikipedia.org/wiki/Browser_Helper_Object
- Malwarebytes Labs, "Browser Hijack Objects (BHOs)" — malwarebytes.com/blog/threats/browser-hijack-objects-bhos
- eWeek, "Enough with the Browser Toolbars Already," Larry Seltzer, Feb 10, 2009 — eweek.com/security/enough-with-the-browser-toolbars-already/

Scenario 2 — Platform lockdown follows abuse (Firefox XUL/XPCOM → WebExtensions, 2015–2017)

Firefox's legacy extension system (XUL/XPCOM) gave add-ons near-unrestricted access to browser internals. In August 2015 Mozilla announced it would deprecate this model in favor of a restricted, Chrome-compatible WebExtensions API, citing security vulnerabilities from unrestricted internal access, and incompatibility with Firefox's move to a multiprocess architecture. The transition (completed with Firefox 57 in late 2017) broke thousands of extensions; many were rewritten, others were abandoned because their functionality could not be replicated under the new, more restrictive framework. Some users migrated to forks (Pale Moon, Waterfox) specifically to keep legacy extension support.

- Wikipedia, "XPInstall" (Deprecation / Impact sections, cited to BleepingComputer and others)
- LWN.net, "Deprecating XUL for WebExtensions," Nathan Willis, Dec 23, 2015 — lwn.net/Articles/668956
- Mozilla Add-ons Blog / wiki.mozilla.org, WebExtensions FAQ and "Evolution in Add-on Development" archives

Scenario 3 — Trusted extensions bought and converted into adware (2014–2018, repeated pattern)

Multiple independent, documented cases show the same acquisition-and-conversion pattern: "Add to Feedly" (30,000+ users) and "Tweet This Page" were sold by their original developers in 2013–2014 and updated by new owners to inject ads and affiliate links into every site visited. In 2017, "Particle" (formerly YouTube+) was sold to a developer who added an ad-injection module within the same update cycle; the extension was pulled from the Chrome Web Store after user complaints. The popular "Stylish" extension was sold in stages between 2016 and 2017 and was found to be logging users' full browsing history. Separately, one of the developers of the couponing extension Honey stated publicly that the team had been repeatedly approached by malware companies, data-collection firms, and adware companies wanting to buy the extension or its user data — and declined.

- PCWorld / IDG, "Spammers buy Chrome extensions and turn them into adware," Lucian Constantin, Jan 20 2014
- G5 Cyber Security / BleepingComputer, "'Particle' Chrome Extension Sold to New Dev Who Immediately Turns It Into Adware," Jul 13 2017
- The Daily Swig (PortSwigger), "They form part of everyday web surfing — but some add-ons harbor a dark secret" (Stylish case)
- Infosecurity Magazine, "Google Pulls Two Adware Extensions From Chrome" (Honey developer quote)

Scenario 4 — Manifest V3 restricts ad-blocker capability, contested by EFF (2019–2022)

Google proposed Manifest V3 in late 2018, restricting the blocking form of the webRequest API (used by content and ad blockers to intercept and block network requests) in favor of a more limited declarativeNetRequest API with a hard cap on filter rules (30,000 at launch, versus ad-blocker filter lists that commonly exceeded 90,000 entries). The Electronic Frontier Foundation published multiple posts opposing the change ("Manifest V3 is Deceitful and Threatening," 2021), arguing it would not stop malicious extensions but would materially reduce ad- and tracker-blocking capability. Mozilla stated it had no plans to remove blocking webRequest from Firefox. Google reworked the rule limits after the pushback, but EFF staff technologists maintained the changes still imposed unnecessary limits on developers.

- LWN.net, "Content blockers and Chrome's Manifest V3," Jake Edge, Dec 21 2021 — lwn.net/Articles/879063
- EFF Deeplinks, "Chrome Users Beware: Manifest V3 is Deceitful and Threatening," Dec 2021 — eff.org/deeplinks/2021/12/chrome-users-beware-manifest-v3-deceitful-and-threatening
- EFF Deeplinks, "Manifest V3: Open Web Politics in Sheep's Clothing," Nov 2021 — eff.org/deeplinks/2021/11/manifest-v3-open-web-politics-sheeps-clothing
- Neowin, "The EFF will fight Google Chrome Manifest v3 which kills extensions that reliably block ads," Dec 14 2021

Scenario 5 — Peer-reviewed and empirical research confirms systemic permission overreach (2016–2025)

This is not a single incident but a recurring empirical finding across independent research groups. MIT researchers (2016) found Chrome's permission model fails to accurately convey an extension's real capabilities to users. A University of Wisconsin–Madison study found roughly 12.5% of studied extensions could access sensitive data such as plaintext passwords present in site source code, identifying 190 extensions — including some with 10M+ installs — directly accessing password fields. A 2022 peer-reviewed study (Chalmers University) found only 39.8% of analyzed Chrome Web Store extensions complied with the principle of least privilege. A 2025 industry audit (Cybernews, cited by Techlicious) found 86% of the top 100 Chrome extensions request high-risk permissions on install.

- Chipman, Ivonchyk, Man, Voss, "Security Analysis of Chrome Extensions," MIT 6.857, May 2016 — courses.csail.mit.edu/6.857/2016/files/24.pdf
- University of Wisconsin–Madison research, summarized via AdGuard, "Extensions Steal Passwords Chrome"
- "Are Chrome extensions compliant with the spirit of least privilege?", Chalmers University of Technology, 2022 — research.chalmers.se/publication/532102
- Techlicious, "Why Your Browser Extensions Could Be a Privacy Nightmare," Suzanne Kantra, Apr 18 2025 (citing Cybernews study)

Where the Five Perspectives Fight

Point of Disagreement	Who	Positions
Is building an extension the right default move?	Practitioner vs. Skeptic	Practitioner treats it as the standard path. Skeptic argues a bookmarklet, web app, or userscript is often the better call — sharpest disagreement in the set.
Are today's APIs a stable foundation to build on?	Practitioner vs. Historian & Skeptic	Practitioner treats the current API surface as workable. Historian and Skeptic contradict this at the root, citing V2→V3 as proof Google's platform control determines outcomes.
How serious is the Manifest V3 transition?	Practitioner vs. Academic	Practitioner frames it as developer inconvenience. Academic frames it as a substantive, documented policy harm with real losers — a disagreement in scope, not a flat contradiction.
Is "cheap to build, cheap to distribute" a genuine advantage?	Economist vs. Skeptic	Economist presents low cost as a real structural pro. Skeptic implicitly discounts it — platform dependency risk outweighs low entry cost.

Strongest to Weakest Evidence, by Voice

Rank	Voice	Basis
1	Academic	Grounded in citable empirical research — permission-overreach studies, content-script attack-surface papers, the V3 policy debate. Most falsifiable claims.
2	Historian	Rests on specific, checkable historical events (IE toolbars/BHOs, Firefox XUL→WebExtensions, Chrome V2→V3). Pattern-inference from history rather than direct measurement.
3	Practitioner	Reliable experiential testimony, verifiable by anyone who builds one — but testimony, not data.
4	Economist	Logically coherent (incentives, acquisition-for-ad-injection pattern, now evidenced in Scenario 3) but more inferential on the broader monetization claim.
5	Skeptic	Weakest evidentially — mostly framing and opinion. Its conclusions get partial, independent backing from the Historian and Academic, but the Skeptic's own argument doesn't supply that evidence.

What All Five Agree On

- **Extensions grant unusually broad, trusted access** — each voice touches this from a different angle: wide API surface, platform risk from that power, data-collection potential, the power-before-lockdown pattern, and documented permission overreach.
- **Google's platform control is the dominant constraint**, not the developer's own choices — even the Practitioner treats Chrome Web Store review delays as a fact of life rather than something within a developer's control.

Coverage Map

Topic	Addressed by	Notably silent
Manifest V3 transition	Practitioner, Skeptic, Historian, Academic	Economist
Security / privacy risk	Skeptic, Economist, Academic (explicit); Historian (implicit)	Practitioner
Monetization / business model	Economist	Everyone else
Cross-platform historical precedent	Historian	Everyone else
Day-to-day build mechanics	Practitioner	Everyone else

Notable gap: the Economist never addresses Manifest V3 at all, despite it being the most-discussed concrete event across the other four — a real blind spot, since V3 materially affected ad-blocker monetization.

Bias Check on This Synthesis

An earlier pass over-weighted the Historian and Academic (three of five findings traced back to their claims), compressing the Economist's and Practitioner's more ground-level, decision-relevant points into supporting roles. This brief corrects for that by restoring the Economist's monetization finding to a central position and downgrading confidence on findings that had relied on unverified "cited research" framing rather than traceable sources.

The Hidden Connection

Trust is the actual product being built — and the actual thing being exploited. No single perspective states this directly, but it underlies all five:

- The Practitioner's easy build works *because* Chrome grants trusted access with little friction.
- The Skeptic's platform-risk argument exists *because* that trust is Google's to grant or revoke, not the developer's.
- The Economist's acquisition pattern (Scenario 3) exists *because* buying a trusted install base is cheaper than earning one from scratch.
- The Historian's lockdown cycle (Scenarios 1, 2, 4) exists *because* that trust gets abused at scale, forcing the platform to claw capability back.
- The Academic's security findings (Scenario 5) exist *because* users grant extensions more access than they grant random websites, and that gap is the attack surface.

Every pro and every con in this brief traces back to one variable: **user trust, monetized as browser access.** The extension isn't really the product. The access is.

Recommended Action

Before building, write down your monetization path independent of the extension itself — then minimize permissions to match that path, not your feature wishlist.

Permission minimization alone treats platform risk as the only problem. The Economist's finding says the business-model risk comes first, chronologically: without a monetization plan beyond "the extension is the product," permission minimization optimizes the wrong variable. Decide the business model first; let that decide the permission scope; treat the resulting permission list as your risk register, not your starting point. This is the one action that responds to findings from all five perspectives at once.

Method note: the five "voices" (Practitioner, Skeptic, Economist, Historian, Academic) are an analytical framing device used to organize this brief, not verbatim quotations from named individuals. The scenario references above are real, independently verifiable sources gathered via web search to support the historical and empirical claims made through those voices.